



SmartHub 2.0

Output Specification

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Table of Contents

SmartHub 2.0 Output Specification	3
Revision History	3
1. Introduction	3
1.1. Report File creation	3
1.2. Configuration files and hierarchy	4
1.3. Translations	4
1.4. CSV Global Options	9
1.5. Converter ("Options")	9
2. Cycle / Process data (C file)	12
2.1. Structure	12
3. Header data (K file)	14
3.1. Structure	14
3.2. Supported column names	16
4. Material data (M file)	18
4.1. Structure	18
4.2. Supported column names	19
5. Endoscopy Single Channel Check data (E file)	20
5.1. Structure	20
5.2. Supported column names	21
6. PDF File	22
6.1. Charts configuration	22
6.2. Table configuration	23
6.3. Batch Content configuration	24
6.4. Printing	24
7. Table of Units	26
8. Appendix	29
8.1. C file default configuration	29
8.2. K file default configuration	35
8.3. M file default configuration	43
8.4. E file default configuration	45
8.5. PDF file default configuration	46

SmartHub 2.0 Output Specification

This document specifies the SmartHub 2.0 output files and its configuration.

Revision History

Table 1. Document Revision

Revision	Date	Author
14.8.3	2026-01-20	Belimed AG

1. Introduction

The SmartHub Connect connects to Belimed devices (sterilizers and cleaning machines). The raw data generated by these machines will be mapped to target files (C,K,M,E, and PDF output files) to be handed over to the ITS (Instrument Tracking System).

- C file = Cycle/Process data
- K file = Header data
- M file = Material data
- E file = Endoscopy Single Channel Check data
- PDF file

1.1. Report File creation

As soon as the device signals processing of a washing/sterilizing cycle is complete, and the device-delivered DCL file (detailed cycle log) or sensor data telegrams are available, the creation of the C file starts.



Orion/iQ devices collect sensor data during a washing/sterilizing cycle on their own and provide the result as a downloadable DCL file in CSV format for download.
For "legacy" devices, SmartHub Connect collects sensor data periodically querying the device during a cycle and storing the gathered sensor data in a local DCL file in CSV format.

When the device signals the batch/cycle is released, the batch information is written to a bmd.json file (bmd = Batch Meta Data) and the creation of the K, M, E and PDF files starts (creation of each file type can be activated/deactivated).

Once all the configured output files have been created, they will all be moved to the target reports folder (the folder ITS is looking at) at once. The C, K, M, E files are stored in CSV format (comma separated value), the PDF file in a pdf format. Finally, the DCL and/or batch.json files are zipped and kept for further inspection.

Values for C file and PDF-table content are taken from the device-delivered DCL (detailed cycle log) files, or from sensor data telegrams.

Values for K, M and E files are gathered from various information sources (configs, device- and batch-metadata).

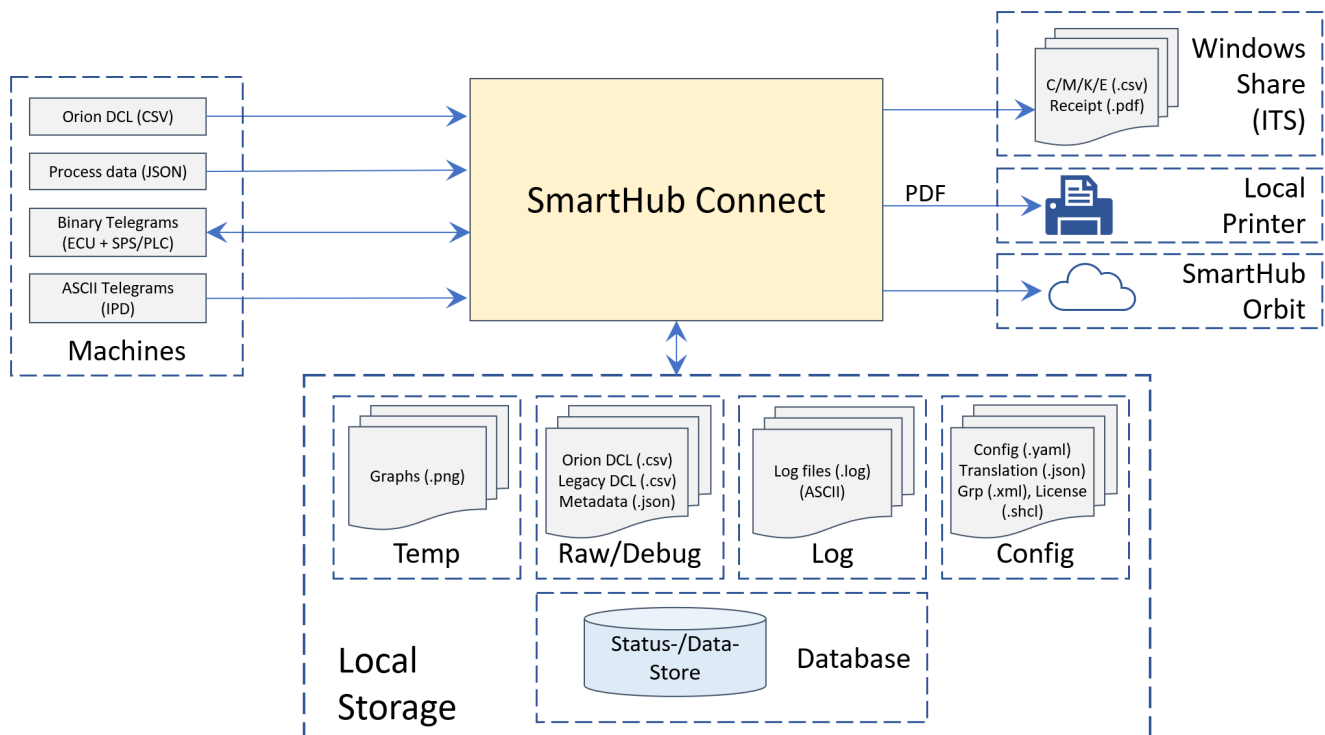


Figure 1. Data and File Formats

1.2. Configuration files and hierarchy

Each output file has a specific configuration file (e.g. `configKFile.yaml`). It contains the mapping rules about how to generate the output files.

These mapping rules can be defined on three different hierarchical levels:

- Globally valid
- Valid for one or more groups of Belimed devices
- Valid for one or more specific Belimed devices (identified by their serial number/equipment number)

Rules defined for a specific device overrule group-level definitions. And group-level definitions overrule global definitions. Therefore, global definitions apply to all machines that have neither a group nor a specific device definition. Mappings for unknown serialNumbers are ignored (no error).

Rules are not merged across levels. I.e. if there is a partial rule definition for either a group or a specific device, no rules are inherited from the upper level definition, but the defaults for these rules apply instead.

Each file contains (on any level) an `activated` property, which when set to true activates creation of the respective target file.

1.3. Translations

Several languages are configured for the output files. Currently, SmartHub Connect supports English, German, French, Spanish and Chinese translations.

These translations are stored in the `english.json`, `german.json`, `french.json`, `spanish.json` and `chinese.json` files.

For the C,K,M,E files, the language used is configured by the `language` property. For the PDF files, it is configured by the `pdflanguage` property. If no language is configured, default translations are used (stored in the `default.json` file).

The translations are used for the following:

1.3.1. Phase-translations

Used to translate phase text numbers into cleartext ("legacy" devices only). To reference them, use the `option: 13`-mapper in the `configCFile.yaml`, and set the mapper's `prefix` attribute to `phase-translations`. (notice the dot), and leave the `postfix` attribute empty.

1.3.2. Cycle-result-translations

Used for the K file in-column `FinalCycleStatus` and for the PDF cycle result. The mapping looks by default like:

```
"cycle-result-translations": {
  "0": "0-Cycle Passed: with no alarms",
  "1": "1-Cycle Passed: with non-critical alarms",
  "2": "2-Cycle Failed: with critical alarms",
  "3": "3-Cycle Failed: cycle aborted",
  "5": "5-Cycle terminated undefined due to communication failure"
},
```

The key `cycle-result-translations.5` is used if the result is unknown, not available or if SmartHub Connect missed the end of a cycle ("legacy" devices only).

1.3.3. Test-result-translations

Used for the K-file in-column `Test result`. If the test result key is not contained in the following translation keys 0, 10, 11, 20, 21 the code will be printed out, otherwise the translated value. The mapping looks by default like:

```
"test-result-translations": {
  "0": "No test result",
  "10": "Bowie Dick test failed",
  "11": "Bowie Dick test passed",
  "20": "Batch control failed",
  "21": "Batch control passed"
},
```

1.3.4. Error-translations

The "error-translations" are used to translate error and alarm numbers into cleartext. To reference them, use the "option: 13"-mapper in the `configCFile.yaml`, and set the mapper's `prefix` attribute to `error-translations`. (notice the dot), and leave the `postfix` attribute empty. For Steelco machine types, also "events" are mapped to the "Message" column. The translations for the event codes are located in `event-translations` list. For those machines an `in2` column and a corresponding `prefix2` attribute are used. They will be used as fallback to read and translate values in case the `in` column is empty.

1.3.5. Ekp-result-translations

Used to translate the value of channel result into cleartext (automatically used for E-files, no config necessary in mapping files).

1.3.6. Position-translations

Used to translate position numbers into cleartext (automatically used for M-files, no config necessary in mapping files).

1.3.7. Display-translations

The "display-translations" contain names for HMI display process messages, currently used for Steelco machines only.

1.3.8. Event-translations

The "event-translations" contain "event" names listed in .BIN files downloaded from Steelco machines, currently used for Steelco machines only.

1.3.9. Pdf-translations

The following image shows how the labels in the file are assigned to the parts of the PDF file. Every text that starts with "PDF" or "Chart" represents a label in the translation file:

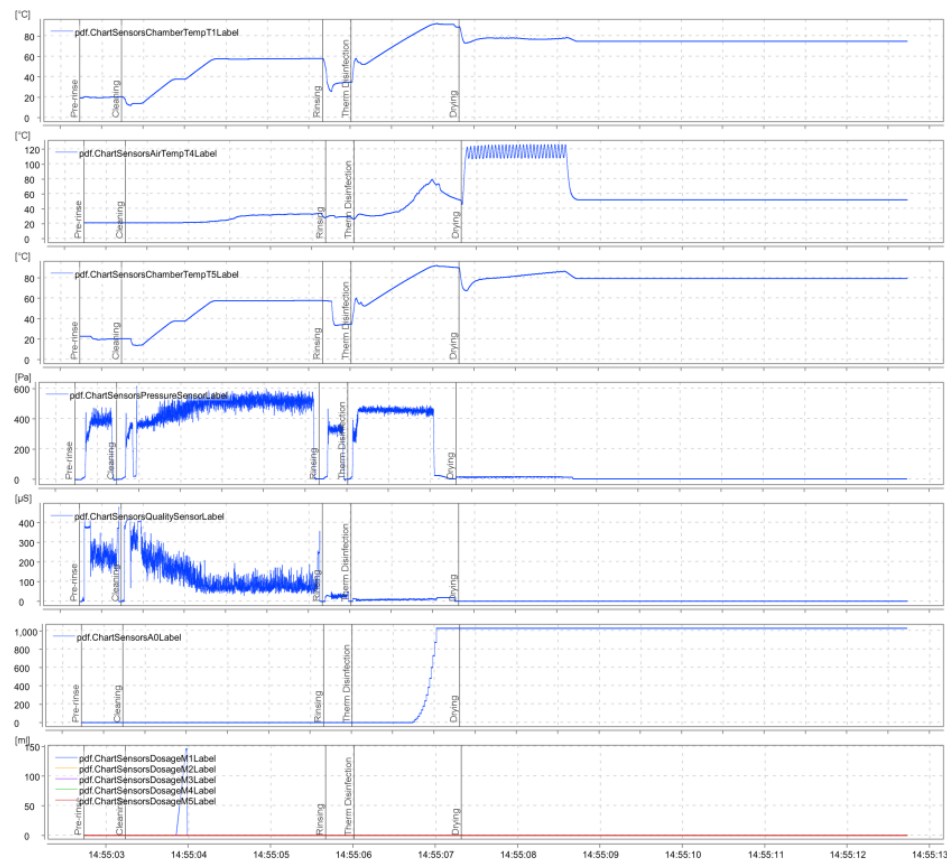
machine 01

pdf.PDFHeaderTop

7

pdf.customerTitle	pdf.machineTitle	pdf.machineAliasTitle	pdf.serialNumberTitle
required	WD290_IQ / V1.6.4	machine 01	10000000001

pdf.batchResultTitle	pdf.programVersionTitle	pdf.progNameTitle	pdf.batchNumberTitle	pdf.cycleStartTitle
Cycle Passed: with no alarms	0 / 1.0.0	P01 Instruments short	7	15.03.2022 14:55:02
pdf.cycleTimeTitle	pdf.disinfTimeTitle	pdf.a0Title	pdf.batchContentTitle	
00:25	125	838	424147	



pdf.operatorTitle	pdf.approval1Title	pdf.approval2Title	pdf.approval3Title
Administrator	pdf.approval1	pdf.approval2	pdf.approval3
pdf.acceptRemTitle			
pdf.signatureTitle			

pdf.PDFPage 1

Figure 2. Translation file label mapping Washer main PDF page

machine 01		pdf.PDFHeaderTop										7
pdf.timestamp	pdf.phase	pdf.messageNumber	pdf.message	CT °C	A0	pdf.waterTemperatureSump	pdf.airTemperatureReleaser	pdf.chamberTemperature	pdf.us	pdf.washingPressure		
14:55:02	Pre-rinse			20.40	0	20.40	21.50	20.40	-1.00	-1.10		
14:55:02	Pre-rinse	PL:176	Validation expired	19.30	0	19.30	21.50	22.90	-0.60	-1.50		
14:55:03	Cleaning			57.90	0.0	57.90	28.00	57.60	-0.60	1.50		
14:55:05	Rinsing			34.80	0.0	34.80	26.10	34.80	-0.40	-0.10		
14:55:06	Therm Disinfection			88.90	1027.34	88.90	51.30	89.90	-0.20	1.50		
14:55:07	Drying			75.00	0.0	75.00	52.30	79.30	-0.20	2.30		
14:55:08	Drying	WD:156	Temperature high	75.70	0.0	75.70	52.80	80.10	-0.20	13.90		

Figure 3. Translation file label mapping Washer table PDF page

1.3.10. Configuration hierarchy of translations

All those files also allow their translations to be defined on three different hierarchical levels:

- Globally valid
- Valid for one or more groups of Belimed devices
- Valid for one or more specific Belimed devices (identified by their serial number)

Translations defined for a specific device overrule group-level translations. And group-level translations overrule global translations.

Unlike `config[C|M|K|E|PDF]File.yaml` files translations *are* merged across levels. I.e. if there is a partial rule definition for either a group or a specific device, translations are still inherited from the upper level definition. The most specific translation given will be taken.

The global configuration must contain all PDF-labels. All other translations are optional.

1.4. CSV Global Options

The following globally applicable CSV options are available in the `configCustomer.yaml` config file.

Option under <code>csv</code> node	default	allowed values	Remark
<code>separator</code>	;	any ASCII string	Separator between columns
<code>usecrlf</code>	true	true, false	Use CRLF "\r\n" for line endings (true) or LF "\n" (false)
<code>format</code>	UTF-8	UTF-8, Windows1252, ISO8859_15	Specifies the encoding of the file
<code>headerInDoubleQuotes</code>	false	true, false	Put column header names in double quotes (") (true) or not (false)
<code>stringsInDoubleQuotes</code>	false	true, false	Put column containing string values in double quotes (") (true) or not (false). If false is selected, then still values that require escaping are enclosed in double quotes
<code>decimalSeparator</code>	.	"," , ","	Decimal point symbol used for decimal numbers

The settings are applicable to CMKE-CSV files only. The raw DCL files that are also stored, are not affected by these settings. The configured `decimalSeparator` is also used in the "PDF table" (last page on the PDF).

1.5. Converter ("Options")

To configure individual entries, specific converters (or "options") are available. A maximum of one option can be added to each entry.

For the C and PDF file all 14 options can be used. For the K, M and E files, options 1 and 2 can be used.

Find below a list of all available converters (options) and their configuration possibilities:

```
# Option 1:
# TimeStamp converter
attribute:
  option: 1
  timeFormatIn: "2006-01-02T15:04:05.000"
  timeFormatOut: "15:04:05.000"
# set option 1
# set time format for input value
# set time format for Output value

# Option 2:
# Set a constant string for the output
attribute:
  option: 2
  constant: "WD"
# set option 2
# The output will be always "WD"

# Option 3:
```

```

# Connect strings with separators (in + separator + in2 + separator2 + in3)
attribute:
  option: 3                                # set option 3
  separator: "."                            # set first separator
  in2: "vA0"                               # set second string from DetailedCycleLog
  separator2: " - "                        # set second separator
  in3: "vDosageM3"                         # set third string from DetailedCycleLog

# Option 4:
# Get the highest float number from in and in2
attribute:
  option: 4
  in2: "aiT4TempAir"

# Option 5
# Convert in2 to an integer and add 1 to it, then connect strings
attribute:
  option: 5
  in2: "smIntTest1"
  separator: "."
  in3: "vDosageM3"
  separator2: " - "

# Option 6
# Reset value after step change
attribute:
  option: 6
  in2: "StageIndex"
  defaultOutput: "0.0"

# Option 7
# Merge of different columns
attribute:
  option: 7
  in2: "StageIndex" # Value 2
  in3: "StageIndex" # Value 3
  in4: "StageIndex" # Value 4
  in5: "StageIndex" # Value 5
  in6: "StageIndex" # Phase index for change detection
  ignoreValue: "0.0"
  defaultOutput: "0"

# Option 8
# Get label from not empty column
attribute:
  option: 8
  label: "StageIndex"
  in2: "StageIndex"
  label2: "StageIndex"
  in3: "StageIndex"
  label3: "StageIndex"
  in4: "StageIndex"
  label4: "StageIndex"
  in5: "StageIndex"
  label5: "StageIndex"
  in6: "StageIndex" # Phase index for change detection
  ignoreValue: "0.0"
  defaultOutput: "0"

# Option 9
# Actual value cannot be lower than the old value except the phase number has changed
attribute:
  option: 9
  in2: "StageIndex" # Phase index for change detection

# Option 10 + 11 are not supported for SmartHub Connect 2.0

# Option 12
# Set chamber Sensor for different stage names
attribute:
  option: 12
  in: "" # Default chamber sensor
  in2: "StageCaption" # Row stage name
  ctSensorMapping:

```

```

- stageName: "Vorspühlen"
  sensor: "aiT1TempChamb"
- ...

# Option 13
# Lookup translation for a given value. The translation key used for lookup is a concatenation
# of <prefix><in value><postfix> without any additional separators. If no value found in the column "in", look if there is
# one in the column "in2". For values from "in2", "prefix2" is used for translations lookup.
# Values from the 'in2' column must be single-valued (not comma separated) and must not be repeated (every occurrence is printed)
# in2, prefix, prefix2, and postfix are optional. The language looked up is defined on the filemapping level.
  attribute:
    option: 13
    in: Error
    in2: Event # optional
    out: Message
    prefix: phase-translations.
    prefix2: event-translations. # optional
    postfix: ""
    ignoreValue: "0" # optional, ignore the given 'in' value (do output "")

# Option 14
# Map alarms to the column. If more than one alarm is contained in the source column, only the first will be written
# to the output, and the others are kept back and are written to the next rows, one by one. Any given alarm is written
# to the output only once per C/PDF-file. I.e. if alarms are contained repeatedly in the DCL, they will still not be
# written repeatedly to the C/PDF-file. This rule applies to values from the 'in' column only. Values that stem from the
# 'in2' column are written everytime they occur. If the 'in' column is empty, the 'in2' column (if present) is also queried.
# Values from the 'in2' column must be single-valued (not comma separated) and must not be repeated (every occurrence is printed)
  attribute:
    option: 14
    in: Error
    in2: Event
    filterIn2: 12,14-16,117,144 # optional, filter in2-values with ids (single or range, comma separated)
    out: Message
    ignoreValue: "0" # optional, ignore the given 'in' value (do output "")
    separator: ", " # default is ,

# Take an input string, split it by the given separatorIn pattern, and rejoin it using the given separator string.
# If the output format is XML, then the separator is used as tag name for child elements.
  attribute:
    option: 15
    in: Sieves
    out: BatchSummary
    separatorIn: " | " # Default is ", "
    separator: "Set" # this example only makes sense if the output is XML. Default is ", "

# Take an input value, recode it using the mapping table. Values without mapping are passed through as they are
  attribute:
    option: 16
    in: CycleResult
    out: ProcessResult
    recode:
      - from: 0
        to: 1
      - from: 3
        to: 2
      - from: 5
        to: 3

```

The file-specific configurations are described in the following chapters [C file](#), [K file](#), [M file](#), [E file](#), and [PDF file](#).

2. Cycle / Process data (C file)

The first line of this CSV file contains the column designations. The next lines contain the measurement values (process data) of the configured sensors.

The number of exported lines depends on the duration of the cycle and the defined sample interval (depends on machine's control unit).

The number format of the measurement values and the units are defined in the table of units (see chapter *Table of units*).

2.1. Structure

The following table shows the global mapping.

In	Out	Option	Description	Type
"Machine Number"	"Maschinen Nr."		Machine no.	num
"Batch Counter"	"Lfd. Chargen-Nr."		Consecutive cycle no.	num
"Timestamp"	"Uhrzeit"	1	Time of captured record. Format depends on regional settings of operating system.	chr[8]
"StageIndex"	"Schritt"	5	Process step (empty in case of sterilizer).	chr[5]
"StageCaption"	"Phase"		Name of process phase.	chr[40]
"Error"	"Meld. Nr."	14	Alarm / fault number.	num
"Error"	"Meldung"	13	Name of alarm / fault (translated).	chr[40]
"do1ColdWater"	"Medium"		Measurement / process value of probe 1	num
"aiT1TempChamb"	"CT °C"	12	Measurement / process value of probe 2	num
"vDosageM1"	"DoV_ml"	7	...	num
"vDosageM1"	"DNo"	8		num
"vA0"	"A0"	6		num
"aiT1TempChamb"	"Wassertemperatur Sumpf"			num
"aiT4TempAir"	"Lufttemperatur Heizung"			num
"aiT5TempChambCtrl"	"Kammertemperatur"			num
"aiC37SensCndctPrb"	"uS"			num

In	Out	Option	Description	Type
"aiP33Pressure"	"Waschdruck"			num
"vDosageM1"	"Dosierpumpe 1"	9		num
"vDosageM2"	"Dosierpumpe 2"	9		num
"vDosageM3"	"Dosierpumpe 3"	9		num
"vDosageM4"	"Dosierpumpe 4"	9		num
"vDosageM5"	"Dosierpumpe 5"	9	Measurement / process value of probe n	num

The C file configuration contains an optional [dataRowsIntervalMillis](#) property. The purpose of the property is to write data rows in the interval provided by this property (at every *x* milliseconds).

Valid values are between *0* and *300000ms*. There is no default value. If not specified, all data rows will be written to the C file.

If there are less data rows available in the data received from the machine than this interval property demands, then at max. the available rows are written.

Data that contains [errors](#), [phase changes](#) (last row of a phase and first row of a phase) will always be written to the file, no matter the configuration value. For Orion/iQ-devices, this property is not supported.

For more information/examples on configuring this file, see the [configCFile.yaml](#).

3. Header data (K file)

The CSV header file always consists of two lines (first line containing the column designations, second line containing the corresponding data of a cycle).

3.1. Structure

The following table shows the global mapping.

In	Out	Option	Description	Type
"Machine.SerialNumber"	"Maschine Nr."		Machine no.	num
"BatchNumber"	"Lfd. Chargin-Nr."		Consecutive cycle no.	num
"UserName"	"Benutzer"		User / operator who started the cycle.	chr[20]
"Started"	"Programmstart"		Timestamp of cycle start. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
"Stopped"	"Programmende"		Timestamp of cycle end. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
"FinalCycleStatus"	"Programm Durchgang"		Status of cycle at process end (code and text). Possible status texts: "0-Cycle Passed: with no alarms", "1-Cycle Passed: with non-critical alarms", "2-Cycle Failed: with critical alarms", "3-Cycle Failed: cycle aborted", "4-Cycle in progress", "5-Cycle has communication alarm, end undefined"	chr[80]
"ProgramNr"	"Programm Nr."		Selected program number at cycle start.	num
"ProgramName"	"Programm Name"		Name of the program.	chr[20]
"ProgramVersion"	"Programm Version"		Program version.	chr[10]
"Machine.FirmWareVersion"	"Software Index"		Version of control unit's software.	chr[6]
"BatchTime"	"Chargendauer"		Duration of the cycle in [sec].	num
"DisinfectionTime"	"Dauer Desinfektion"		Period of sterilization or disinfection in [sec]. The column designation depends on the type of machine.	num
"MinSterilizationTemp"	"Min. Sterilisiertemp."		Minimum sterilization temperatur in [°C].	num
"MaxSterilizationTemp"	"Max. Sterilisiertemp."		Maximum sterilization temperatur in [°C].	num

In	Out	Option	Description	Type
"F0"	"F0-Wert"		Calculated F0 value in [min].	num
"LeakRate"	"Leckrate"		Calculated leak rate in [mbar / min].	num
"LastBowieDickTest"	"Letzter B&D-Test"		Timestamp of previous Bowie-Dick Test. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
"LastBowieDickTestBatchNo"	"Letzter B&D-Test Lfd. Nr."		Cycle no. of previous Bowie-Dick Test.	num
"LastVacuumTest"	"Letzter Vakuumtest"		Timestamp of previous vacuum leak test. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
"LastVacuumTestBatchNo"	"Letzter Vakuumtest Lfd. Nr."		Cycle no. of previous vacuum leak test.	num
""	"Funktionskontrolle"		Function control. (applies to cleaning machines of type WD280 and WD380 with B&R control unit)	chr[1]
""	"Chargenfreigabe(Name)"		Name of user of electronic cycle / batch approval.	chr[40]
""	"Chargenfreigabe(Datum/Uhrzeit)"		Timestamp of electronic cycle / batch approval. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Chargenfreigabe(Status)"		Status text of electronic cycle / batch approval. Possible status texts: "granted", "denied"	chr[20]
""	"Chargenkenndaten_1"		Batch characteristics data 1 for cycle.	chr[40]
""	"Chargenkenndaten_2"		Batch characteristics data 2 for cycle.	chr[40]
""	"Chargenkenndaten_3"		Batch characteristics data 3 for cycle.	chr[40]
"Machine.Name"	"Maschinen Name"		Name of machine.	chr[20]
"TestResult"	"Testergebnis"		Result text of integrated Bowie-Dick test or batch control. Possible result texts: "Bowie-Dick Test failed", "Bowie-Dick Test passed", "Batch control failed", "Batch control passed"	chr[40]
"IntegratedError1"	"RIF 1"		Integrated error 1.	num
"IntegratedError2"	"RIF 2"		Integrated error 2.	num

In	Out	Option	Description	Type
"IntegratedError3"	"RIF 3"		Integrated error 3.	num

The K file configuration contains an [operatorName](#) property. The purpose of the property is to specify if the operator name shall be included in the K files.

Valid values are [include](#) and [exclude](#). Default value is [include](#).

3.2. Supported column names

The following "in" column names are supported for K files (case-sensitive!):

(They can also be used for the C-file to be referenced by the [in](#) column. Further columns like [in2](#), [in3](#), ... do *not* support them. If there are sensors from the DCL with the same name, those sensors take precedence over the batch metadata values.)

- A0
- A0Setpoint
- BatchId
- BatchNumber
- BatchTime
- Data
- DisinfectionOk
- DisinfectionTime
- F0
- FinalCycleStatus
- FinishedStatus
- HasDetailedLog
- LeakRate
- Machine.Alias
- Machine.FirmWareVersion
- Machine.LastBatchId
- Machine.LastBatchNumber
- Machine.machineId
- Machine.MachineGroup
- Machine.MachineType
- Machine.Name
- Machine.OrderNumber
- Machine.SerialNumber
- Machine.SiteEquipId
- Machine.Url
- MaxSterilizationTemp
- MinSterilizationTemp

- ProcessingStatus
- ProgramId
- ProgramName
- ProgramNr
- ProgramVersion
- RackID
- Sieves
- Started
- SterilizationTime
- Stopped
- TestResult
- IntegratedError1
- IntegratedError2
- IntegratedError3
- LastBowieDickTest
- LastBowieDickTestBatchNo
- LastVacuumTest
- LastVacuumTestBatchNo
- Timestamp
- UserCode (Steelco only)
- UserName
- WorkOrderCode (Steelco only)

For more information/examples on configuring this file, see the [configKFile.yaml](#).

4. Material data (M file)

Each exported line contains the captured material ID (scanned by barcode reader or supplied as reference no. by instrument management system).

4.1. Structure

The following table shows the global mapping.

In	Out	Option	Description	Type
"Machine.SerialNumber"	"Maschine Nr."		Machine no.	num
"BatchNumber"	"Lfd. Chargen-Nr."		Consecutive cycle no.	num
"VpE-Nr."	"VpE-Nr."		System-internal ID of packaging unit (VpE). Always contains a consecutive ID of the packaging unit (VpE). This ID is unique and is automatically generated by the system for each washing or sterilization process.	num
""	"VpE-ID-Vater"		ID of "parent" (formerly used for group units).	num
"Typ"	"Typ"		Possible types: "VpE": General material without base data, "WG": Washing material with base data, "CW": Carrier, cart, washing tray	chr[3]
"Nr."	"Nr."		Material ID (article no., washing material, reference no. in handshake file). Always contains the unique material ID.	chr[40]
""	"Bezeichnung"		Name of material (washing material).	chr[40]
""	"Verfallsdatum"		Date & time of expiration of packaging unit. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Packer"		Name of operator who packed the material.	chr[20]
""	"Verpackdat."		Date & time of packaging. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Einlagerdat."		Date & time of sterile storage (end of cycle). Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Auslagerdat."		Date & time of removing from storage area. Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Auslagerer"		Name of operator who removed the material from storage area.	chr[20]
""	"Verw.ort"		Place of usage for basic unit (BU).	chr[20]

In	Out	Option	Description	Type
""	"Anzahl BEs"		Number of basic units.	num
""	"Patient.-Name" ¹⁾		Name of patient. ¹⁾	chr[40]
""	"Patient.-Nr." ¹⁾		Patient ID. ¹⁾	chr[20]
""	"Reinigungs-Nr."		Cleaning no.	chr[20]
"Timestamp"	"Beladedat."	1	Date & time of loading (batch contents message received from machine). Format: YYYY-MM-DD-hh.mm.ss.nnnnnn	chr[26]
""	"Position"		Position of material (washing material) on wash tray as text in the current ICS system language.	chr[40]

1) "Patient.-Name" and "Patient.-Nr." fields are required to be present by some ITS. However, they always remain empty. SmartHub Connect does never store or handle patient data.

4.2. Supported column names

The following "in" column names are supported for M files (case-sensitive!):

- "Machine.SerialNumber"
- "BatchNumber"
- "VpE-Nr."
- "Typ"
- "Nr."
- "Timestamp"
- "Position"

For more information/examples on configuring this file, see the [configMFile.yaml](#).

5. Endoscopy Single Channel Check data (E file)

This output file is only supported for the WD 430 devices.

This CSV file contains the results of the Single Channel Checking (SCC) which applies to endoscopy machines (WD 430), if these results are transmitted to the ICS at the end of the cleaning process. For each endoscopy channel one line containing the result of the check is exported.

5.1. Structure

The following table shows the global mapping.

In	Out	Option	Description	Type
"Maschinen Nr."	"Maschinen Nr."		Machine no.	num
"Lfd. Chargen-Nr."	"Lfd. Chargen-Nr."		Consecutive cycle no.	num
"Messart"	"Messart"		Measuring mode: 0: Normal, 1: Measure rinse 2, 2: Measure rinse 3	num
"Messart_Txt"	"Messart_Txt"		Text of measuring mode: "F": Normal, "K": Measure rinse 2, "L": Measure rinse 3	chr[1]
"Kanal_Nr"	"Kanal_Nr"		Channel no.: 1 ... 48	num
"Kanal_Resultat"	"Kanal_Resultat"		Result code: 1 ... 255 (derzeit 0 ... 8)	num
"Kanal_Resultat_Txt"	"Kanal_Resultat_Txt"		Result as text (see table below).	chr[40]
"Kanal_Messwert"	"Kanal_Messwert"		Measurement value: 0 ... 999 (Error = -1)	num
"Kanal_Sollwert"	"Kanal_Sollwert"		Set value: 0 ... 999	num
"Kanal_Quelle"	"Kanal_Quelle"		Source of measurement value: 0: ECU- control device, 1: IPD module	num
"Kanal_Quelle_Txt"	"Kanal_Quelle_Txt"		Source of measurement value as text: "ECU", "IPD"	chr[3]

5.1.1. Kanal_Resultat_Txt

Result	Result text	Description
0	No Check	No check performed, value is only displayed
1	Not used	Channel not used, channel deactivated
2	NoConnec	Channel not connected (>+80% or > 950)

Result	Result text	Description
3	Leakage	Leakage
4	O.K.	Passed
5	Failed	Failed
6	NoConnec	Albarran, not connected (≥ 500)
7	O.K.	Albarran passed (< 500)
8	Failed	Albarran failed (sensor failure)

5.2. Supported column names

The following "in" column names are supported for E files (case-sensitive!):

- "Maschinen Nr"
- "Lfd. Chargen-Nr."
- "Messart"
- "Messart_Txt"
- "Kanal_Nr"
- "Kanal_Resultat"
- "Kanal_Resultat_Txt"
- "Kanal_Messwert"
- "Kanal_Sollwert"
- "Kanal_Quelle"
- "Kanal_Quelle_Txt"

For more information/examples on configuring this file, see the [configEFile.yaml](#).

6. PDF File

The PDF file configuration contains an `operatorName` property. The purpose of the property is to specify if the operator name shall be included in the PDF files.

Valid values are `include` and `exclude`. Default value is `include`.

With the optional `reportsFont` property it is possible to configure a custom font for PDF creation. If the property is not set or the configured font file does not exist, the default font (Courier) will be used instead.

6.1. Charts configuration

The `configPDFFile.yaml` file allows to configure what charts are rendered to the PDF by specifying the `charts` property tree. If no `charts` property is present, default charts according to SmartHub 1.0 implementation are rendered. The default charts configuration is as follows:

```
charts:
  - sensors:
    - sensor: "aiT1TempChamb"
      legend: "ChartSensorsChamberTempT1Label"
      yaxis: left1
    - sensor: "aiT4TempAir"
      legend: "ChartSensorsAirTempT4Label"
      yaxis: left1
    - sensor: "aiT5TempChambCtrl"
      legend: "ChartSensorsChamberTempT5Label"
      yaxis: left1
  yaxes:
    left1:
      caption: "[°C]"
  marker:
    columnName: "StageCaption"
  height: 420
  - sensors:
    - sensor: "aiP33Pressure"
      legend: "ChartSensorsPressureSensorLabel"
      yaxis: left1
  yaxes:
    left1:
      caption: "[hPa]"
  marker:
    columnName: "StageCaption"
  - sensors:
    - sensor: "aiC37SensCndctPrb"
      legend: "ChartSensorsQualitySensorLabel"
      yaxis: left1
  yaxes:
    left1:
      caption: "[μS]"
  marker:
    columnName: "StageCaption"
  - sensors:
    - sensor: "vA0"
      legend: "ChartSensorsA0Label"
      yaxis: left1
  yaxes:
    left1:
      caption: "StageCaption"
  marker:
    columnName: "StageCaption"
  - sensors:
    - sensor: "vDosageM1"
      legend: "ChartSensorsDosageM1Label"
      yaxis: left1
    - sensor: "vDosageM2"
      legend: "ChartSensorsDosageM2Label"
      yaxis: left1
    - sensor: "vDosageM3"
      legend: "ChartSensorsDosageM3Label"
      yaxis: left1
```

```

- sensor: "vDosageM4"
  legend: "ChartSensorsDosageM4Label"
  yaxis: left1
- sensor: "vDosageM5"
  legend: "ChartSensorsDosageM5Label"
  yaxis: left1
yaxes:
  left1:
    caption: "[ml]"
marker:
  columnName: "StageCaption"

```

6.2. Table configuration

It is configurable whether the PDF contains a table of sampled data. The PDF table typically contains entries marking the start and end of each phase, along with sensor values at those times. In addition, a separate entry is added for every alarm that occurs.

A `table` element contains an `activated` property, which when set to `true` activates creation of the table in the PDF. The `filemapping` property in the `table` section describes the columns in the table. It can be used as described above. The table can display properly maximum 15 columns.

The `dataInterval` property in the `table` section allows to define time intervals at which extra rows are added to the table. This property accepts a list of interval objects, each of which must include the required `phase` and `intervalSeconds` fields:

- `phase` specifies the phase number during which the additional data rows should be written.
- `intervalSeconds` defines the time interval, in seconds, at which these rows are added.

`intervalSeconds` must not be set to a value lower than 10 seconds.

The phase number refers to the DCL, namely values from columns according to the following rules:

- IQ machines: column `StageIndex`
- MST machines: column `PhaseTextNum`, or `StageCaption` if `PhaseTextNum` is null
- Belimed legacy washers: column `PhaseTextNum`

The phase number corresponds to DCL values, determined from specific columns depending on the machine type:

- **IQ machines:** `StageIndex` column
- **MST machines:** `PhaseTextNum` column, or `StageCaption` if `PhaseTextNum` is null
- **Belimed legacy washers:** `PhaseTextNum` column

```

dataInterval:
- phase: 1
  intervalSeconds: 20
- phase: 4
  intervalSeconds: 10

```

The global table configuration is shown below:

In	Out	Option	Description	Type
"Timestamp"	"Uhrzeit"	1	Time of captured record. Format depends on regional settings of operating system.	chr[8]
"StageCaption"	"Phase"		Name of process phase.	chr[40]
"Error"	"Meld. Nr."	14	Alarm / fault number.	num
"Error"	"Meldung"	13	Name of alarm / fault (translated)	chr[40]
"aiT1TempChamb"	"CT °C"	12	Measurement / process value of probe 2	num
"vA0"	"A0"	6		num
"aiT1TempChamb"	"Wassertemperatur Sumpf"			num
"aiT4TempAir"	"Lufttemperatur Heizung"			num
"aiT5TempChambCtrl"	"Kammertemperatur"			num
"aiC37SensCndctPrb"	"uS"			num
"aiP33Pressure"	"Waschdruck"			num

6.3. Batch Content configuration

The `configPDFFile.yaml` file allows to configure if the batch content page is shown, by specifying the `batchContent` property tree.

If `batchContent` is not present, the following default settings apply:

```
batchContent:
  activated: true
  showMaterial: true
  showEkpResults: simple # none | simple | detailed
```

6.4. Printing

It is configurable whether the PDF is immediately sent to the default printer by setting the property `printer` to `true`.

By default, this feature uses the default printer configured for the user running the Windows Service `SmartHub`. It is possible to configure the name of another printer by defining the optional yaml property `printerName`. If the printer configured by `printerName` is not found, the Windows user's default printer is used instead.

If the property `printerName` is missing or empty, or if the printer is not found, and if no default printer is configured for the Windows user, then an error message is written to the log and the PDF file will not be printed.

Example:

```
printer: true  
printerName: "HP OfficeJet Pro 7740 series"
```

For more information/examples on configuring this file, see the [configPDFFile.yaml](#).

7. Table of Units

These units or physical values as well as the presentation of actual measurement values are defined for all equipments:

Unit-ID	Unit	Presentation	Decimal point	Transmission format*)	Symbol (remarks)
1	°C	N3.1	1	°C/10 (deciDegree)	T (temperature)
2	°F	N3.1	1	°C/10 (deciDegree)	T (temperature)
3	mbar a	N4.0	0	mbar a	P (absolute)
4	mbar r	N4.0	0	mbar a	P (relative)
5	bar a	N1.3	3	mbar a	P (absolute)
6	bar r	N1.3	3	mbar a	P (relative)
7	kPa a	N3.1	1	mbar a	P (absolute)
8	kPa r	N3.1	1	mbar a	P (relative)
9	psig	N2.2	2	mbar a P	(excess pressure)
10	in.Hg	N2.2	2	mbar a	P (vacuum)
11	µS/cm	N4.0	0	µS/cm	"Gamma" (conductivity)
12	ml/l	N3.1	1	ml/l /10 (deci-ml/l)	D (dosage)
13	sec	N4.0	0	sec	t (time)
14	min	N3.1	1	min/10 (deciminute)	t (time)
15	mbar/min	N4.0	0	mbar/min	R ramp "Delta" P/"Delta" t
16	(0)	N2.0	0	nn	(unit or utility number)
17	(1)	N2.1	1	nnn	(e.g. step no.)
18	V	N3.1	1	V/10 (decivolt)	U (voltage)
19	ml	N3.1	1	ml/10	Dosage
20	A	N2.1	1	A/10 (deciampere)	I (current)
21	psia	N2.2	2	mbar a	P (absolute pressure)
28	g	N4.2	2	1/100g	m (mass/weight)

Unit-ID	Unit	Presentation	Decimal point	Transmission format*)	Symbol (remarks)
30	%	N3.2	2	1/100%	Percent
32	Pa a	N6.0	0	mbar a	P (absolute)
33	Pa r	N6.0	0	mbar a	P (relative)
34	mS/cm	N4.1	2	µS/cm	"Gamma" (conductivity)
35	Char(20)	Char(20)	-	ASCII	freely definable
36	ppm	N5.0	0	Parts per million	
37	ppb	N5.0	0	Parts per billion	
38	µS/cm(2)	N4.2	2	µS/cm/100	"Gamma" (conductivity)
39	torr	N4.0	0	mbar a	P (absolute pressure)
Steelco specific units (often different scaling factor compared to Belimed version)					
200	l	N1.1	1	l/10 (deciLiter)	V (volume)
201	ml	N4.0	0	ml	V (volume)
202	bar	N2.2	2	bar/100 (centiBar)	P (absolute pressure)
203	µS	N3.1	1	µS/10	"Gamma" (conductivity)
204	pH	N3.1	1	pH/10	pH
205	seconds	N3.0	0	seconds	t (time)
206	minutes	N3.0	0	minutes	t (time)
207	%	N2.0	0	%	percentage
208	%	N2.1	1	%/10	percentage
209		N2.2	2	1/100	no unit
210	mV	N3.2	2	mV/10	U (voltage)
211	l/min	N5.0	0	l/min	Q (flow rate)
212	l	N5.0	0	l	V (volume)
213	rpm	N5.0	0	rpm	Rounds per minute

Unit-ID	Unit	Presentation	Decimal point	Transmission format*)	Symbol (remarks)
214	W	N5.0	0	W	P (power)
215	mm/s	N5.0	0	mm/s	v (speed)
216	p/min	N5.0	0	p/min	pulses/min

*) The measurement values of the sensors are captured from the control devices by using the internal transmission format (signed integer).

8. Appendix

The following chapters lists the default configurations for each machine group per file type.

8.1. C file default configuration

8.1.1. Global C file config

```

activated: true
language: german
filemapping:
  - in: "Machine Number"
    out: "Maschinen Nr."
  - in: "Batch Counter"
    out: "Lfd. Chargen-Nr."
  - in: "Timestamp"
    out: "Uhrzeit"
    option: 1
    timeFormatIn: "2006-01-02T15:04:05.000"
    timeFormatOut: "15:04:05"
  - in: "StageIndex"
    out: "Schritt"
    option: 5
    separator: "."
    in2: "smIntTest1"
    in2cut: 3
  - in: "StageCaption"
    out: "Phase"
  - in: "Error"
    out: "Meld. Nr."
    option: 14
    separator: ","
    ignoreValue: "0"
  - in: "Error"
    out: "Meldung"
    option: 13
    ignoreValue: "0"
    prefix: "error-translations."
  - in: "do1ColdWater"
    label: "1"
    in2: "do2WarmWater"
    label2: "2"
    in3: "do3VeWater"
    label3: "3"
    in4: "do4VeTankEmpty"
    label4: "4"
    in5: "do5PTankEmpty"
    in6: "smIntTest1"
    label5: "5"
    option: 8
    ignoreValue: "0"
    defaultOutput: "0"
    out: "Medium"
  - in: "aiT1TempChamb"
    in2: "StageCaption"
    out: "CT °C"
    option: 12
    ctSensorMapping:
      - stageName: "Vorspühler"
        sensor: "aiT1TempChamb"
      - stageName: "Trocknen"
        sensor: "aiT1TempChamb"
  - in: "vDosageM1"
    in2: "vDosageM2"
    in3: "vDosageM3"
    in4: "vDosageM4"
    in5: "vDosageM5"
    in6: "smIntTest1"
    option: 7
    ignoreValue: "0.0"
    defaultOutput: "0.0"

```

```
out: "DoV_ml"
- in: "vDosageM1"
  label: "1"
  in2: "vDosageM2"
  label2: "2"
  in3: "vDosageM3"
  label3: "3"
  in4: "vDosageM4"
  label4: "4"
  in5: "vDosageM5"
  in6: "smIntTest1"
  label5: "5"
  option: 8
  ignoreValue: "0.0"
  defaultOutput: "0"
  out: "DNo"
- in: "vA0"
  out: "A0"
  option: 6
  in2: "StageIndex"
  defaultOutput: "0.0"
- in: "aiT1TempChamb"
  out: "Wassertemperatur Sumpf"
- in: "aiT4TempAir"
  out: "Lufttemperatur Heizung"
- in: "aiT5TempChambCtrl"
  out: "Kammertemperatur"
- in: "aiC37SensCndetPrb"
  out: "uS"
- in: "aiP33Pressure"
  out: "Waschdruck"
- in: "vDosageM1"
  out: "Dosierpumpe 1"
  option: 9
  in2: "smIntTest1"
- in: "vDosageM2"
  out: "Dosierpumpe 2"
  option: 9
  in2: "smIntTest1"
- in: "vDosageM3"
  out: "Dosierpumpe 3"
  option: 9
  in2: "smIntTest1"
- in: "vDosageM4"
  out: "Dosierpumpe 4"
  option: 9
  in2: "smIntTest1"
- in: "vDosageM5"
  out: "Dosierpumpe 5"
  option: 9
  in2: "smIntTest1"
```

8.1.2. WD150_2XX, WD430 C file config

```
- groupName:
  - "WD150_2XX"
  - "WD430"
language: german
activated: true
filemapping:
  - in: "Machine Number"
    out: "Maschinen Nr."
  - in: "Batch Counter"
    out: "Lfd. Chargen-Nr."
  - in: "Timestamp"
    out: "Uhrzeit"
    option: 1
    timeFormatIn: "2006-01-02T15:04:05"
    timeFormatOut: "15:04:05"
  - in: "StageIndex"
    out: "Schritt"
  - in: "PhaseTextNum"
    out: "Phase"
    option: 13
    prefix: phase-translations.
  - in: "Error"
    out: "Meld. Nr."
    option: 14
    separator: ","
    ignoreValue: "0"
  - in: "Error"
    out: "Meldung"
    option: 13
    prefix: error-translations.
    ignoreValue: "0"
  - in: "Med"
    out: "Medium"
  - in: "CT"
    out: "CT °C"
  - in: "DNo1"
    out: "CDNo1"
  - in: "DoV1"
    out: "DosV ml"
  - in: "A0"
    out: "Ao"
  - in: "Zonennr"
    out: "Zonen-Nr."
    option: 2
    constant: "1"
```


8.1.3. WD750 C file config

```
- groupName: "WD750"
activated: true
language: german
filemapping:
  - in: "Machine Number"
    out: "Maschinen Nr."
  - in: "Batch Counter"
    out: "Lfd. Chargen-Nr."
  - in: "Timestamp"
    out: "Uhrzeit"
    option: 1
    timeFormatIn: "2006-01-02T15:04:05"
    timeFormatOut: "15:04:05"
  - in: "StageIndex"
    out: "Schritt"
  - in: "PhaseTextNum"
    out: "Phase"
    option: 13
    prefix: phase-translations.
  - in: "Error"
    out: "Meld. Nr."
    option: 14
    separator: ", "
    ignoreValue: "0"
  - in: "Error"
    out: "Meldung"
    option: 13
    prefix: error-translations.
  - in: "MED"
    out: "Medium"
  - in: "TWII"
    out: "TWII °C"
  - in: "TTRI"
    out: "TTRI °C"
  - in: "PI"
    out: "PI bar a"
```

8.1.4. MST C file config

```
- groupName:
  - "MST"
  - "MST_TOP5000"
activated: true
language: german
filemapping:
  - in: "Machine Number"
    out: "Maschinen Nr."
  - in: "Batch Counter"
    out: "Lfd. Chargen-Nr."
  - in: "Timestamp"
    out: "Uhrzeit"
    option: 1
    timeFormatIn: "2006-01-02T15:04:05"
    timeFormatOut: "15:04:05"
  - in: "StageIndex"
    out: "Schritt"
  - in: "PhaseTextNum"
    out: "Phase"
    option: 13
    prefix: phase-translations.
  - in: "Error"
    out: "Meld. Nr."
    option: 14
    separator: ","
    ignoreValue: "0"
  - in: "Error"
    out: "Meldung"
    option: 13
    prefix: error-translations.
  - in: "T1"
    out: "T1 °C"
  - in: "T2"
    out: "T2 °C"
  - in: "T3"
    out: "T3 °C"
  - in: "T4"
    out: "T4 °C"
  - in: "P1"
    out: "P1 mbar a"
  - in: "P2"
    out: "P2 mbar a"
  - in: "P3"
    out: "P3 mbar a"
  - in: "P4"
    out: "P4 mbar a"
  - in: "F01"
    out: "F01 min"
  - in: "F02"
    out: "F02 min"
  - in: "Zonennr"
    out: "Zonen-Nr."
    option: 2
    constant: "1"
```

8.1.5. CS750 C file config

```
- groupName: "CS750"
  activated: true
  language: german
  filemapping:
    - in: "Machine Number"
      out: "Maschinen Nr."
    - in: "Batch Counter"
      out: "Lfd. Chargen-Nr."
    - in: "Timestamp"
      out: "Uhrzeit"
      option: 1
      timeFormatIn: "2006-01-02T15:04:05"
      timeFormatOut: "15:04:05"
    - in: "StageIndex"
      out: "Schritt"
    - in: "PhaseTextNum"
      out: "Phase"
      option: 13
      prefix: phase-translations.
    - in: "Error"
      out: "Meld. Nr."
      option: 14
      separator: ", "
      ignoreValue: "0"
    - in: "Error"
      out: "Meldung"
      option: 13
      prefix: error-translations.
    - in: "WTU"
      out: "WTU °C"
    - in: "STU"
      out: "STU °C"
    - in: "TRK1"
      out: "TRK1 °C"
    - in: "TRK2"
      out: "TRK2 °C"
    - in: "WI"
      out: "WI °C"
    - in: "DoNr"
      out: "DoNr"
    - in: "DoMe"
      out: "DoMe ml"
    - in: "A0"
      out: "A0"
    - in: "Zonennr"
      out: "Zonen-Nr."
      option: 2
      constant: "1"
```

8.2. K file default configuration

8.2.1. Global K file config

```

activated: true
language: german
operatorName: include
filemapping:
  - in: "Machine.SerialNumber"
    out: "Maschine Nr."
  - in: "BatchNumber"
    out: "Lfd. Chargen-Nr."
  - in: "UserName"
    out: "Benutzer"
  - in: "Started"
    out: "Programmstart"
  - in: "Stopped"
    out: "Programmende"
  - in: "FinalCycleStatus"
    out: "Programm Durchgang"
  - in: "ProgramNr"
    out: "Programm Nr."
  - in: "ProgramName"
    out: "Programm Name"
  - in: "ProgramVersion"
    out: "Programm Version"
  - in: "Machine.FirmWareVersion"
    out: "Software Index"
  - in: "BatchTime"
    out: "Chargendauer"
  - in: "DisinfectionTime"
    out: "Dauer Desinfektion"
  - in: "MinSterilizationTemp"
    out: "Min. Sterilisiertemp."
  - in: "MaxSterilizationTemp"
    out: "Max. Sterilisiertemp."
  - in: "F0"
    out: "F0-Wert"
  - in: "LeakRate"
    out: "Leckrate"
  - in: ""
    out: "Letzter B&D-Test"
  - in: ""
    out: "Letzter B&D-Test Lfd. Nr."
  - in: ""
    out: "Letzter Vakuumtest"
  - in: ""
    out: "Letzter Vakuumtest Lfd. Nr."
  - in: ""
    out: "Funktionskontrolle"
  - in: ""
    out: "Chargenfreigabe(Name)"
  - in: ""
    out: "Chargenfreigabe(Datum/Uhrzeit)"
  - in: ""
    out: "Chargenfreigabe(Status)"
  - in: ""
    out: "Chargenkenndaten_1"
  - in: ""
    out: "Chargenkenndaten_2"
  - in: ""
    out: "Chargenkenndaten_3"
  - in: "Machine.Name"
    out: "Maschinen Name"
  - in: ""
    out: "Testergebnis"
  - in: ""
    out: "RIF 1"
  - in: ""
    out: "RIF 2"
  - in: ""
    out: "RIF 3"

```

8.2.2. WD150_2XX, WD430, WD750 K file config

```

- groupName:
  - "WD150_2XX"
  - "WD430"
  - "WD750"
language: german
activated: true
operatorName: include
filemapping:
  - in: "Machine.SerialNumber"
    out: "Maschine Nr."
  - in: "BatchNumber"
    out: "Lfd. Chargen-Nr."
  - in: "UserName"
    out: "Benutzer"
  - in: "Started"
    out: "Programmstart"
  - in: "Stopped"
    out: "Programmende"
  - in: "FinalCycleStatus"
    out: "Programm Durchgang"
  - in: "ProgramNr"
    out: "Programm Nr."
  - in: "ProgramName"
    out: "Programm Name"
  - in: "ProgramVersion"
    out: "Programm Version"
  - in: "Machine.FirmWareVersion"
    out: "Software Index"
  - in: "BatchTime"
    out: "Chargendauer"
  - in: "DisinfectionTime"
    out: "Dauer Desinfektion"
  - in: ""
    out: "Min. Sterilisiertemp."
  - in: ""
    out: "Max. Sterilisiertemp."
  - in: ""
    out: "F0-Wert"
  - in: ""
    out: "Leckrate"
  - in: ""
    out: "Letzter B&D-Test"
  - in: ""
    out: "Letzter B&D-Test Lfd. Nr."
  - in: ""
    out: "Letzter Vakuumtest"
  - in: ""
    out: "Letzter Vakuumtest Lfd. Nr."
  - in: ""
    out: "Funktionskontrolle"
  - in: ""
    out: "Chargenfreigabe(Name)"
  - in: ""
    out: "Chargenfreigabe(Datum/Uhrzeit)"
  - in: ""
    out: "Chargenfreigabe(Status)"
  - in: ""
    out: "Chargenkenndaten_1"
  - in: ""
    out: "Chargenkenndaten_2"
  - in: ""
    out: "Chargenkenndaten_3"
  - in: "Machine.Name"
    out: "Maschinen Name"
  - in: ""
    out: "Testergebnis"
  - in: ""
    out: "RIF 1"
  - in: ""
    out: "RIF 2"
  - in: ""

```

out: "RIF 3"

8.2.3. MST K file config

```

- groupName:
  - "MST"
  - "MST_TOP5000"
language: german
activated: true
operatorName: include
filemapping:
  - in: "Machine.SerialNumber"
    out: "Maschine Nr."
  - in: "BatchNumber"
    out: "Lfd. Chargen-Nr."
  - in: "UserName"
    out: "Benutzer"
  - in: "Started"
    out: "Programmstart"
  - in: "Stopped"
    out: "Programmende"
  - in: "FinalCycleStatus"
    out: "Programm Durchgang"
  - in: "ProgramNr"
    out: "Programm Nr."
  - in: "ProgramName"
    out: "Programm Name"
  - in: "ProgramVersion"
    out: "Programm Version"
  - in: "Machine.FirmWareVersion"
    out: "Software Index"
  - in: "BatchTime"
    out: "Chargendauer"
  - in: "SterilizationTime"
    out: "Dauer Sterilisation"
  - in: "MinSterilizationTemp"
    out: "Min. Sterilisiertemp."
  - in: "MaxSterilizationTemp"
    out: "Max. Sterilisiertemp."
  - in: "F0"
    out: "F0-Wert"
  - in: "LeakRate"
    out: "Leckrate"
  - in: "LastBowieDickTest"
    out: "Letzter B&D-Test"
  - in: "LastBowieDickTestBatchNo"
    out: "Letzter B&D-Test Lfd. Nr."
  - in: "LastVacuumTest"
    out: "Letzter Vakuumtest"
  - in: "LastVacuumTestBatchNo"
    out: "Letzter Vakuumtest Lfd. Nr."
  - in: ""
    out: "Funktionskontrolle"
  - in: ""
    out: "Chargenfreigabe(Name)"
  - in: ""
    out: "Chargenfreigabe(Datum/Uhrzeit)"
  - in: ""
    out: "Chargenfreigabe(Status)"
  - in: ""
    out: "Chargenkenndaten_1"
  - in: ""
    out: "Chargenkenndaten_2"
  - in: ""
    out: "Chargenkenndaten_3"
  - in: "Machine.Name"
    out: "Maschinen Name"
  - in: "TestResult"
    out: "Testergebnis"
  - in: "IntegratedError1"
    out: "RIF 1"
  - in: "IntegratedError2"
    out: "RIF 2"
  - in: "IntegratedError3"

```

out: "RIF 3"

8.2.4. CS750 K file config

```

- groupName:
  - "CS750"
  language: german
  activated: true
  operatorName: include
  filemapping:
    - in: "Machine.SerialNumber"
      out: "Maschine Nr."
    - in: "BatchNumber"
      out: "Lfd. Chargen-Nr."
    - in: "UserName"
      out: "Benutzer"
    - in: "Started"
      out: "Programmstart"
    - in: "Stopped"
      out: "Programmende"
    - in: "FinalCycleStatus"
      out: "Programm Durchgang"
    - in: "ProgramNr"
      out: "Programm Nr."
    - in: "ProgramName"
      out: "Programm Name"
    - in: "ProgramVersion"
      out: "Programm Version"
    - in: "Machine.FirmWareVersion"
      out: "Software Index"
    - in: "BatchTime"
      out: "Chargendauer"
    - in: "DisinfectionTime"
      out: "Dauer Desinfektion"
    - in: ""
      out: "Min. Sterilisiertemp."
    - in: ""
      out: "Max. Sterilisiertemp."
    - in: ""
      out: "F0-Wert"
    - in: ""
      out: "Leckrate"
    - in: ""
      out: "Letzter B&D-Test"
    - in: ""
      out: "Letzter B&D-Test Lfd. Nr."
    - in: ""
      out: "Letzter Vakuumtest"
    - in: ""
      out: "Letzter Vakuumtest Lfd. Nr."
    - in: ""
      out: "Funktionskontrolle"
    - in: ""
      out: "Chargenfreigabe(Name)"
    - in: ""
      out: "Chargenfreigabe(Datum/Uhrzeit)"
    - in: ""
      out: "Chargenfreigabe(Status)"
    - in: ""
      out: "Chargenkenndaten_1"
    - in: ""
      out: "Chargenkenndaten_2"
    - in: ""
      out: "Chargenkenndaten_3"
    - in: "Machine.Name"
      out: "Maschinen Name"
    - in: ""
      out: "Testergebnis"
    - in: ""
      out: "RIF 1"
    - in: ""
      out: "RIF 2"
    - in: ""
      out: "RIF 3"

```

8.2.5. K file config for Asanus ITS

```

activated: true
operatorName: include
fileFormat: XML # default is CSV
filePrefix: "" # default is "K_"
fileExtension: .BCDCR # default is .xml for XML format and .csv for CSV format
xmlRootElem: BatchInformations # only used if fileFormat is XML
filemapping:
  - in: "Machine.OrderNumber"
    out: "MachineNumber"
    format: "00" # must be enclosed in "" !
  - in: "Machine.alias"
    out: "MachineName"
  - in: ""
    out: "MachineType"
    constant: "2"
    option: 2
  - in: "BatchNumber"
    out: "BatchNumber"
  - in: "ProgramNr"
    out: "ProgramNumber"
    format: "00"
  - in: "ProgramName"
    out: "ProgramName"
  - in: "Started"
    out: "StartDateTime"
    option: 1
    timeFormatOut: 20231231235959
  - in: "Stopped"
    out: "EndDateTime"
    option: 1
    timeFormatOut: 20231231235959
  - in: "UserName"
    out: "UserStart"
  - in: "CycleResult"
    out: "ProcessResult"
    option: 16
    recode:
      - from: 0
        to: 1
      - from: 3
        to: 2
      - from: 5
        to: 3
  - in: ""
    out: "PDFFileNameWithPath"
    option: 3
    in2: "PDFFileNameWMachinePrefix"
    separator: "C:\\pdf\\root\\path\\as\\seen\\by\\Asanus\\" # SERVICE: This needs to be adjusted based on customer setup
  - in: "RackID"
    out: "WasherTrolleyBarcode"
  - in: "Sieves"
    out: "BatchSummary"
    option: 15
    separatorIn: " | "
    separator: Set

groups:
  - groupName:
    - "MST"
    - "MST_F0"
    - "MST_TOP5000"
  activated: true
  operatorName: include
  fileFormat: XML # default is CSV
  filePrefix: "" # default is "K_"
  fileExtension: .BCDCR # default is .xml for XML format and .csv for CSV format
  xmlRootElem: BatchInformations # only used if fileFormat is XML
  filemapping:
    - in: "Machine.OrderNumber"
      out: "MachineNumber"
      format: "00"

```

```
- in: "Machine.alias"
  out: "MachineName"
- in: ""
  out: "MachineType"
  constant: "1"
  option: 2
- in: "BatchNumber"
  out: "BatchNumber"
- in: "ProgramNr"
  out: "ProgramNumber"
  format: "00"
- in: "ProgramName"
  out: "ProgramName"
- in: "Started"
  out: "StartDateTime"
  option: 1
  timeFormatOut: 20231231235959
- in: "Stopped"
  out: "EndDateTime"
  option: 1
  timeFormatOut: 20231231235959
- in: "UserName"
  out: "UserStart"
- in: "CycleResult"
  out: "ProcessResult"
  option: 16
  recode:
    - from: 0
      to: 1
    - from: 3
      to: 2
    - from: 5
      to: 3
- in: ""
  out: "PDFFileNameWithPath"
  option: 3
  in2: "PDFFileNameWMachinePrefix"
  separator: "C:\\pdf\\root\\path\\as\\seen\\by\\Asanus\\" # SERVICE: This needs to be adjusted based on customer setup
  # No rackID and sieves for sterilizers
```

8.3. M file default configuration

8.3.1. Global M file config

```
activated: true
language: english
filemapping:
  - in: "Machine.SerialNumber"
    out: "Maschine Nr."
  - in: "BatchNumber"
    out: "Lfd. Chargen-Nr."
  - in: "VpE-Nr."
    out: "VpE-Nr."
  - in: ""
    out: "VpE-ID-Vater"
  - in: "Typ"
    out: "Typ"
  - in: "Nr."
    out: "Nr."
  - in: ""
    out: "Bezeichnung"
  - in: ""
    out: "Verfallsdatum"
  - in: ""
    out: "Packer"
  - in: ""
    out: "Verpackdat."
  - in: ""
    out: "Einlagerdat."
  - in: ""
    out: "Auslagerdat."
  - in: ""
    out: "Auslagerer"
  - in: ""
    out: "Verw.ort"
  - in: ""
    out: "Anzahl BEs"
  - in: ""
    out: "Patient.-Name"
  - in: ""
    out: "Patient.-Nr."
  - in: ""
    out: "Reinigungs-Nr."
  - in: "Timestamp"
    out: "Beladedat."
  option: 1
  timeFormatIn: "2006-01-02T15:04:05"
  timeFormatOut: "2006-01-02"
  - in: ""
    out: "Position"
```

8.3.2. WD430 M file config

```
- groupName:  
  - "WD430"  
activated: true  
language: english  
filemapping:  
  - in: "Machine.SerialNumber"  
    out: "Maschine Nr."  
  - in: "BatchNumber"  
    out: "Lfd. Chargen-Nr."  
  - in: "VpE-Nr."  
    out: "VpE-Nr."  
  - in: ""  
    out: "VpE-ID-Vater"  
  - in: "Typ"  
    out: "Typ"  
  - in: "Nr."  
    out: "Nr."  
  - in: ""  
    out: "Bezeichnung"  
  - in: ""  
    out: "Verfallsdatum"  
  - in: ""  
    out: "Packer"  
  - in: ""  
    out: "Verpackdat."  
  - in: ""  
    out: "Einlagerdat."  
  - in: ""  
    out: "Auslagerdat."  
  - in: ""  
    out: "Auslagerer"  
  - in: ""  
    out: "Verw.ort"  
  - in: ""  
    out: "Anzahl BEs"  
  - in: ""  
    out: "Patient.-Name"  
  - in: ""  
    out: "Patient.-Nr."  
  - in: ""  
    out: "Reinigungs-Nr."  
  - in: "Timestamp"  
    out: "Beladedat."  
option: 1  
timeFormatIn: "2006-01-02T15:04:05"  
timeFormatOut: "2006-01-02"  
  - in: "Position"  
    out: "Position"
```

8.4. E file default configuration

8.4.1. Global E file config

```
activated: false
```

8.4.2. WD430 E file config

```
- groupName:
  - "WD430"
  activated: true
  language: german
  filemapping:
    - in: "Maschinen Nr."
      out: "Maschinen Nr."
    - in: "Lfd. Chargen-Nr."
      out: "Lfd. Chargen-Nr."
    - in: "Messart"
      out: "Messart"
    - in: "Messart_Txt"
      out: "Messart_Txt"
    - in: "Kanal_Nr"
      out: "Kanal_Nr"
    - in: "Kanal_Resultat"
      out: "Kanal_Resultat"
    - in: "Kanal_Resultat_Txt"
      out: "Kanal_Resultat_Txt"
    - in: "Kanal_Messwert"
      out: "Kanal_Messwert"
    - in: "Kanal_Sollwert"
      out: "Kanal_Sollwert"
    - in: "Kanal_Quelle"
      out: "Kanal_Quelle"
    - in: "Kanal_Quelle_Txt"
      out: "Kanal_Quelle_Txt"
```

8.5. PDF file default configuration

8.5.1. Global PDF file config

```
activated: true
pdflanguage: "english"
printer: false
operatorName: include
table:
  activated: true
  filemapping:
    - in: "Timestamp"
      out: "Uhrzeit"
      option: 1
      timeFormatIn: "2006-01-02T15:04:05.000"
      timeFormatOut: "15:04:05"
    - in: "StageCaption"
      out: "Phase"
    - in: "Error"
      out: "Meld. Nr."
      option: 14
      separator: ","
      ignoreValue: "0"
    - in: "Error"
      out: "Meldung"
      option: 13
      prefix: error-translations.
      ignoreValue: "0"
    - in: "aiT1TempChamb"
      in2: "StageCaption"
      out: "CT °C"
      option: 12
      ctSensorMapping:
        - stageName: "Vorspühler"
          sensor: "aiT1TempChamb"
        - stageName: "Trocknen"
          sensor: "aiT1TempChamb"
    - in: "vA0"
      out: "A0"
      option: 6
      in2: "StageIndex"
      defaultOutput: "0.0"
    - in: "aiT1TempChamb"
      out: "Wassertemperatur Sumpf"
    - in: "aiT4TempAir"
      out: "Lufttemperatur Heizung"
    - in: "aiT5TempChambCtrl"
      out: "Kammertemperatur"
    - in: "aiC37SensCndctPrb"
      out: "uS"
    - in: "aiP33Pressure"
      out: "Washdruck"
```

8.5.2. WD150_2XX, WD430 PDF file config

```

- groupName:
  - "WD150_2XX"
  - "WD430"
activated: true
printer: false
pdflanguage: "english"
operatorName: include
charts:
  - sensors:
    - sensor: "CT"
      legend: "CT °C"
      yaxis: left1
    - sensor: "A0"
      legend: "A0"
      yaxis: right1
  yaxes:
    left1:
      caption: "[°C]"
    right1:
      caption: ""
  marker:
    columnName: "PhaseTextNum"
  height: 350
table:
  activated: true
  filemapping:
    - in: "Timestamp"
      out: "Time"
      option: 1
      timeFormatIn: "2006-01-02T15:04:05"
      timeFormatOut: "15:04:05"
    - in: "Error"
      out: "Meld. Nr."
      option: 14
      ignoreValue: "0"
    - in: "Error"
      out: "Meldung"
      option: 13
      prefix: error-translations.
      ignoreValue: "0"
    - in: "PhaseTextNum"
      out: "Phase"
      option: 13
      prefix: phase-translations.
    - in: "CT"
      out: "CT °C"
    - in: "A0"
      out: "A0"

```


8.5.3. WD750 PDF file config

```

- groupName: "WD750"
  activated: true
  printer: false
  pdflanguage: "english"
  operatorName: include
  charts:
    - sensors:
        - sensor: "TWTI"
          legend: "TWTI °C"
          yaxis: left1
        - sensor: "TTTRI"
          legend: "TTTRI °C"
          yaxis: left1
        - sensor: "PI"
          legend: "PI bar a"
          yaxis: right1
      yaxes:
        left1:
          caption: "[°C]"
        right1:
          caption: "bar a"
      marker:
        columnName: "PhaseTextNum"
      height: 350
  table:
    activated: true
    filemapping:
      - in: "Timestamp"
        out: "Time"
        option: 1
        timeFormatIn: "2006-01-02T15:04:05"
        timeFormatOut: "15:04:05"
      - in: "Error"
        out: "Meld. Nr."
        option: 14
        ignoreValue: "0"
      - in: "Error"
        out: "Meldung"
        option: 13
        prefix: error-translations.
        ignoreValue: "0"
      - in: "PhaseTextNum"
        out: "Phase"
        option: 13
        prefix: phase-translations.
      - in: "TWTI"
        out: "TWTI °C"
      - in: "TTTRI"
        out: "TTTRI °C"
      - in: "PI"
        out: "PI bar a"

```

8.5.4. MST PDF file config

```

- groupName:
  - "MST"
  - "MST_TOP5000"
activated: true
printer: false
pdflanguage: "english"
operatorName: include
charts:
  - sensors:
    - sensor: "T1"
      legend: "TWTI °C"
      yaxis: left1
    - sensor: "T2"
      legend: "TTRI °C"
      yaxis: left1
    - sensor: "P1"
      legend: "PI bar a"
      yaxis: right1
  yaxes:
    left1:
      caption: "[°C]"
    right1:
      caption: "bar a"
  marker:
    columnName: "PhaseTextNum"
    height: 350
table:
  activated: true
  filemapping:
    - in: "Timestamp"
      out: "Time"
      option: 1
      timeFormatIn: "2006-01-02T15:04:05"
      timeFormatOut: "15:04:05"
    - in: "Error"
      out: "Meld. Nr."
      option: 14
      ignoreValue: "0"
    - in: "Error"
      out: "Meldung"
      option: 13
      prefix: error-translations.
      ignoreValue: "0"
    - in: "PhaseTextNum"
      out: "Phase"
      option: 13
      prefix: phase-translations.
    - in: "T1"
      out: "TWTI °C"
    - in: "T2"
      out: "TTRI °C"
    - in: "P1"
      out: "PI bar a"

```

8.5.5. CS750 PDF file config

```

- groupName: "CS750"
  activated: true
  printer: false
  pdfLanguage: "english"
  operatorName: include
  charts:
    - sensors:
        - sensor: "WTU"
          legend: "WTU"
          yaxis: left1
        - sensor: "STU"
          legend: "STU"
          yaxis: left1
        - sensor: "TRK1"
          legend: "TRK1"
          yaxis: left1
        - sensor: "TRK2"
          legend: "TRK2"
          yaxis: left1
        - sensor: "WI"
          legend: "WI"
          yaxis: left1
        - sensor: "A0"
          legend: "A0"
          yaxis: right1
      yaxes:
        left1:
          caption: "[°C]"
        right1:
          min: 0
          max: 5000
      marker:
        columnName: "PhaseTextNum"
  table:
    activated: true
    filemapping:
      - in: "Timestamp"
        out: "Time"
        option: 1
        timeFormatIn: "2006-01-02T15:04:05"
        timeFormatOut: "15:04:05"
      - in: "Error"
        out: "Meld. Nr."
        option: 14
        ignoreValue: "0"
      - in: "Error"
        out: "Meldung"
        option: 13
        prefix: error-translations.
        ignoreValue: "0"
      - in: "PhaseTextNum"
        out: "Phase"
        option: 13
        prefix: phase-translations.
      - in: "WTU"
        out: "WTU"
      - in: "STU"
        out: "STU"
      - in: "TRK1"
        out: "TRK1"
      - in: "TRK2"
        out: "TRK2"
      - in: "WI"
        out: "WI"
      - in: "A0"
        out: "A0"

```

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EN English

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